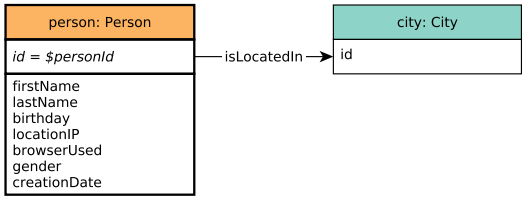
## Interactive / short / 1

IS 1
IS 2
IS 3
IS 4
IS 5
IS 6
IS 7

| | |
|---|---|
| query | Interactive / short / 1 |
| title | Profile of a person |
| pattern |   |
| desc. | Given a start Person, retrieve their first name, last name, birthday, IP address, browser, and city of residence. |
| params | |

| | | | |
|---|---|---|---|
| 1 | personId | ID | |

result

| | | | | |
|---|---|---|---|---|
| 1 | person.firstName | String | R | |
| 2 | person.lastName | String | R | |
| 3 | person.birthday | Date | R | |
| 4 | person.locationIP | String | R | |
| 5 | person.browserUsed | String | R | |
| 6 | city.id | ID | R | |
| 7 | person.gender | String | R | |
| 8 | person.creationDate | DateTime | R | |

## Interactive / short / 2

IS 1
IS 2
IS 3
IS 4
IS 5
IS 6
IS 7

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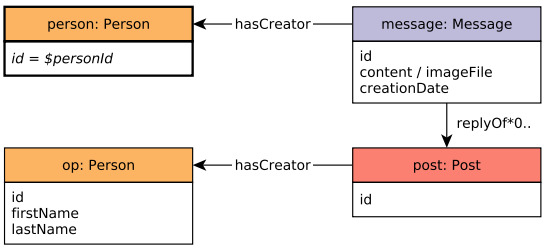
|---|---|
| query | Interactive / short / 2 |
| title | Recent messages of a person |
| pattern |  |
| desc. | Given a start Person, retrieve the last 10 Messages created by that user. For each Message, return that Message, the original Post in its conversation (`post`), and the author of that Post (`originalPoster`). If any of the Messages is a Post, then the original Post (`post`) will be the same Message, i.e. that Message will appear twice in that result. |
| params | |

| | | | |
|---|---|---|---|
| 1 | `personId` | ID | |

result

| | | | | |
|---|---|---|---|---|
| 1 | `message.id` | ID | R | |
| 2 | `message.content` or `message.imageFile` (for photos) | Text | R | |
| 3 | `message.creationDate` | DateTime | R | |
| 4 | `post.id` | ID | R | |
| 5 | `originalPoster.id` | ID | R | |
| 6 | `originalPoster.firstName` | String | R | |
| 7 | `originalPoster.lastName` | String | R | |

sort

| | | | |
|---|---|---|---|
| 1 | `message.creationDate` | ↓ | |
| 2 | `message.id` | ↓ | |

## Interactive / short / 3

IS 1
IS 2
IS 3
IS 4
IS 5
IS 6
IS 7

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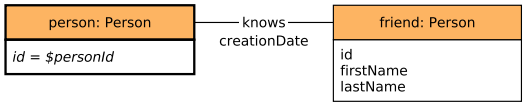
|---|---|
| query | Interactive / short / 3 |
| title | Friends of a person |
| pattern |   |
| desc. | Given a start Person, retrieve all of their friends, and the date at which they became friends. |
| params | 1 `personId` ID |
| result | 1 `friend.id` ID R<br>2 `friend.firstName` String R<br>3 `friend.lastName` String R<br>4 `knows.creationDate` DateTime R |
| sort | 1 `knows.creationDate` ↓<br>2 `friend.id` ↑ |

## Interactive / short / 4

IS 1
IS 2
IS 3
IS 4
IS 5
IS 6
IS 7

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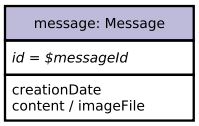
|---|---|
| query | Interactive / short / 4 |
| title | Content of a message |
| pattern |   |
| desc. | Given a Message, retrieve its content and creation date. |
| params | 1 `messageId` ID |
| result | 1 `message.creationDate` DateTime R `messageCreationDate`<br>2 `message.content or message.imageFile (for photos)` Text R `messageContent` |

## Interactive / short / 5

IS 1
IS 2
IS 3
IS 4
IS 5
IS 6
IS 7

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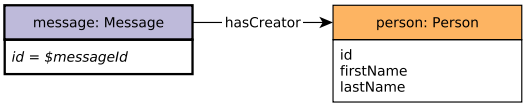
|---|---|
| query | Interactive / short / 5 |
| title | Creator of a message |
| pattern |  |
| desc. | Given a Message, retrieve its author. |
| params | 1 messageId ID |
| result | 1 person.id ID R<br>2 person.firstName String R<br>3 person.lastName String R |

## Interactive / short / 6

IS 1
IS 2
IS 3
IS 4
IS 5
IS 6
IS 7

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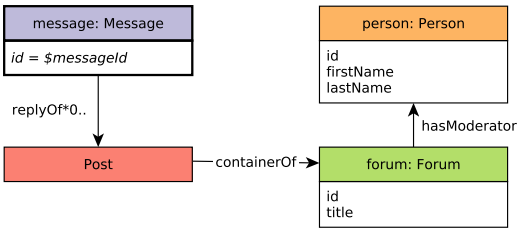
|---|---|
| query | Interactive / short / 6 |
| title | Forum of a message |
| pattern |  |
| desc. | Given a Message, retrieve the Forum that contains it and the Person that moderates that Forum. Since Comments are not directly contained in Forums, for Comments, return the Forum containing the original Post in the thread which the Comment is replying to. |
| params | 1 messageId ID |
| result | 1 forum.id ID R<br>2 forum.title Long String R<br>3 moderator.id ID R<br>4 moderator.firstName String R<br>5 moderator.lastName String R |

## Interactive / short / 7

IS 1
IS 2
IS 3
IS 4
IS 5
IS 6
IS 7

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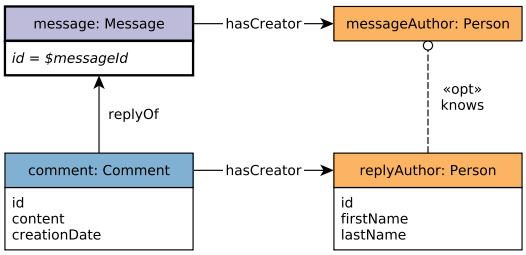
|---|---|
| query | Interactive / short / 7 |
| title | Replies of a message |
| pattern |   |
| desc. | Given a Message, retrieve the (1-hop) Comments that reply to it.<br>In addition, return a boolean flag `knows` indicating if the author of the reply (`replyAuthor`) knows the author of the original message (`messageAuthor`). If author is same as original author, return False for `knows` flag. |
| params | |

| | | | |
|---|---|---|---|
| 1 | `messageId` | ID | |

result

| | | | | |
|---|---|---|---|---|
| 1 | `comment.id` | ID | R | |
| 2 | `comment.content` | Text | R | |
| 3 | `comment.creationDate` | DateTime | R | |
| 4 | `replyAuthor.id` | ID | R | |
| 5 | `replyAuthor.firstName` | String | R | |
| 6 | `replyAuthor.lastName` | String | R | |
| 7 | `knows` | Boolean | C | True if the `knows` edge exists between the `replyAuthor` and the `messageAuthor` nodes, False otherwise (including the case when the two nodes are the same) |

sort

| | | |
|---|---|---|
| 1 | `comment.creationDate` | ↓ |
| 2 | `replyAuthor.id` | ↑ |